



FIT2179 Data Visualisation
Course Notes

DATA TYPES

Monash University

Version 1.1 of 28 July 2025

Based on “Visualization Analysis and Design” (2014) by Tamara Munzner

DATASET TYPES

DATASETS

A dataset is a collection of information. There are four essential dataset types: tables, networks, spatial fields and spatial geometry. These four dataset types are made up of data attributes.

TERMINOLOGY

Synonyms for *attribute* are *variable* and *data dimension* (or just *dimension*). The term dimension has many other meanings; to avoid ambiguity we prefer *attribute* and *variable*.

A dataset is not inherently visible. To visualise a dataset, we select and apply a visualisation method that transforms the invisible dataset into a visual representation. For each dataset type a range of visualisation methods exist. For instance, for table datasets, we can choose from various methods, such as a bar chart, a line chart, or a pie chart. However, not all methods are equally efficient or effective for a given table dataset. Understanding the characteristics of both attributes and dataset types is essential for selecting a visualisation method that ensures the data can be read effectively.

TERMINOLOGY

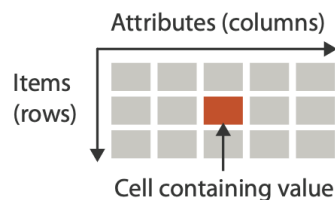
The textbook “Visualization Analysis and Design” by T. Munzner uses visualisation *idiom* instead of visualisation *method*.

TERMINOLOGY

In visualisation, *data* is commonly treated as a singular noun, whereas in the natural sciences, *data* is often a plural noun.

TABLES

A basic table consists of cells with values arranged in rows and columns. Each row represents a data *item*, and each column represents an attribute.



TERMINOLOGY

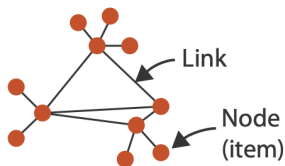
A data *item* (a row in a table) is also called a *record* of a table.

When a table is stored in a database, each item is associated with a key that allows for unambiguous identification and retrieval of that item. A multidimensional table has multiple keys to index each cell. For example, we could build a three-dimensional table storing house prices for different years. The three-dimensional table would have two keys: one key to index the house and one key to index the year.

Columns (attributes) and rows (items) of a table can be reordered, deleted, added, modified, etc. These operations are useful, for example, to sort items in a table by an attribute value, or filter items of interest.

NETWORKS AND TREES

A network encodes the relationships between items. An item in a network is referred to as a *node* or *vertex*. A relationship is called a *link* or *edge*.



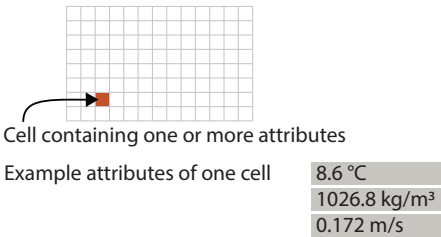
Examples of network datasets include social networks where people are nodes and friendships are links; food webs, where animals are nodes and predator-prey relationships are links; networks of hyperlinks, where webpages are nodes and hyperlinks are links; and air-traffic networks, where airports are nodes and flights are links.

The nodes and links of a network often have attributes. For instance, in an air traffic network, the nodes may have attributes such as the name of the airport and the number of runways, while the links may have attributes such as the number of daily flights and the number of yearly passengers.

A tree is a network without cycles. Each node has only one parent node, and nodes can have multiple child nodes. Examples of trees include file system hierarchies, which organise folders and files; biological taxonomy trees, which classify life forms from kingdoms down to species; and organisational charts, which represent the hierarchy of a company or other institutions.

FIELDS

A field contains values from a continuous spatial domain. The values of a field are measurements or the results of a simulation. Fields typically store values in cells, often square in shape and arranged at a regular interval in a uniform grid. However, more complex forms of fields also exist. For instance, the size of cells can vary to capture more detailed information in specific areas, or square cells can be replaced with irregularly shaped triangles that form a mesh surface.



Fields can have two or three dimensions. A two-dimensional field with square cells is similar to a raster image composed of pixels, but instead of storing colour values, each cell stores attribute values. A three-dimensional field with a regular grid can be imagined as a Rubik's cube with many cells. In analogy to two-dimensional pixels, the cells of a three-dimensional grid are called *voxels*. Voxel models are commonly used in engineering and earth sciences.

The domain of a field is continuous, meaning there are conceptually an infinite number of locations where values could be measured

or simulated. However, a field only stores values at specific, discrete locations. The values can be interpolated to estimate values at positions that do not align with the stored values.

Fields can be categorised into three types based on the number and semantic of their attributes:

- A *scalar field* contains a single attribute.
- A *vector field* contains both directions and magnitudes (typically representing directed speed or force values).
- A *tensor field* contains multiple attributes that logically belong together.

Below are examples of typical field types:

- A CT scan of the interior of a human body is a three-dimensional scalar field, containing measurements of tissue density that indicate how easily X-rays can pass through each voxel.
- A field containing elevation values relative to sea level, sampled using an airborne laser scanner, is a two-dimensional scalar field.
- A field with wind velocities computed by a weather forecast model is a three-dimensional vector field.
- A field encoding landcover information stores the type of landcover at the location of the cells in a two-dimensional grid. For example, 0 could represent water, 1 could represent forest, 2 could represent buildings, etc.

TABLE VS FIELD

Let's explore the differences between tables and fields. A major distinction is that tables are non-spatial, whereas fields are spatial: tables are not positioned within a spatial coordinate system and they do not contain information about the size of their cells. In contrast, fields are defined within a spatial coordinate system, and their cells have specific dimensions. For instance, the three-dimensional cells of a medical CT scan may measure $0.5 \times 0.5 \times 0.5$ mm, while the two-dimensional cells of a surface temperature field for Australia could measure 500×500 m.

A second major difference between tables and fields lies in the operations each dataset type supports.

A table consists of columns (attributes) and rows (items). Each column contains a different attribute, such as *name*, *salary* or *age*. Columns can be reordered, deleted or added. For example, the *salary* and *age* columns could be swapped, and the *name* column could be deleted to anonymise the dataset. Rows (items) can also be reordered, deleted or added. For instance, items could be sorted by decreasing *salary* values.

In short, with tables, operations such as reordering, deleting or adding columns and rows result in a modified dataset that generally remains useful.

This is not the case with fields, which contain samples of a continuous domain, for

example, surface temperature sampled at regularly spaced locations. Fields support spatial analysis; for example, we could calculate the average temperature of a geographic area, or search the highest and lowest temperature inside a specific area.

However, it does not make sense to reorder the rows or columns of a 2D grid, as re-ordering would scramble the grid cells and destroy the spatial continuity of the sampled values. Changing the order or rows or columns would alter the position of cells and make spatial analysis impossible.

In summary, with tables it is permissible to reorder, add or delete rows and columns, whereas doing so with a field would disrupt the spatial continuity of its values and prevent spatial analysis operations.

GEOMETRY

The items in a geometry dataset are points, lines, areas or volumes that are positioned in a spatial coordinate system. Point geometry is dimensionless, representing only a location. Line geometry is one-dimensional, allowing movement along the line in one dimension. Area geometry is two-dimensional, permitting movement along two independent coordinate axes within the area. Volume geometry is three-dimensional, allowing movement along three independent axes within the volume.

Lines, areas and volumes are constructed from points. For example, a line is made up of a string of points, an area is formed from a closed line, and a volume is created from points forming a three-dimensional shape.

Items in a geometry datasets often have attributes. For example, a geometry dataset representing the states and territories of Australia consists of eight items, each with area geometry and attributes such as the state's name and population.

For practical purposes, a geometry dataset often contains only one type of geometry. For example, the 500 most populous cities in the world (only points) or all the lakes in Victoria (only areas). However, a geometry dataset can also contain a mix of different geometry types.

FIELD VS GEOMETRY

Field datasets and geometry datasets are both spatial, meaning they are positioned within a spatial coordinate system. Let's explore the difference between them and how they can be transformed into each other.

A field contains values from a continuous spatial domain. The values may change from one cell to the next, and the level of detail of the information that the field stores depends on its cell size. Consider a satellite image with colours in the spectrum visible to the human eye. If we significantly enlarge this field dataset, we can see the individual pixels (or cells) of the field. When we zoom out, the pixels become indistinguishable and appear as continuous colour gradients.

A dataset with area geometry, however, is different because its shapes create abrupt boundaries. Consider an area geometry representing the eight states and territories of Australia. If each shape is shown with a different colour, zooming in or out always reveals the same eight distinct colours, with borders that remain sharp and clearly defined. This is because the areas are built from strings of point positions.

It is possible to convert a field dataset into a geometry dataset. For instance, we could create area geometries for groups of cells that have identical or similar values.

Conversely, transforming a geometry dataset to a field dataset is also possible. For example, consider measuring soil contamination

with mercury within the area of an industrial complex. For each measurement, we measure a contamination value and also determine the position with a GPS receiver, resulting in a point geometry dataset with a mercury concentration attribute. By applying a spatial interpolation method, we could estimate the mercury concentration throughout the entire area of the industrial complex. The result would be a field dataset, where cells store interpolated mercury concentrations that vary continuously. This transformation from point geometry to a field is useful because it allows us to visualise the spatially continuous nature of mercury contamination with methods specific to field datasets, such as contour lines or colour gradients.

ATTRIBUTE TYPES

There are two main types of data attributes: categorical and ordered. Ordered attributes can be further divided into ordinal and quantitative attributes.

→ Categorical



→ Ordered

→ Ordinal



→ Quantitative



CATEGORICAL ATTRIBUTES

Categorical attributes, also known as *nominal* or sometimes *qualitative* attributes, do not have an inherent order. They differentiate between distinct types of items, each with a unique name (the term *nominal* comes from the Latin *nomen*, meaning “name”).

Examples of categorical attributes include:

- Hair colour (e.g., blonde, brown, black)
- Ethnicity (e.g., Asian, Hispanic)
- Countries (e.g., Mali, Sudan, Chad)
- Car type (e.g., hatchback, sedan, SUV)
- Postal code (e.g., 2000 for Sydney, 3000 for Melbourne)
- Phone number (e.g., 03 9905 4000 for Monash University, 03 9035 5511 for the University of Melbourne)

Note that categorical attributes can be numeric, such as postal codes or phone numbers. However, performing arithmetic operations on these numbers does not make sense.

Some categorical attributes are part of a hierarchy. For example, the house cat (*Felis catus*) belongs to the *Felidae* family, which belongs to the *Carnivora* order. Similarly, Melbourne is a city in the state of Victoria, which is part of Australia.

ORDERED ATTRIBUTES

Ordered attributes are sequential, diverging or cyclic.

Sequential attributes vary between a minimum and a maximum value. Examples include age and depth below sea level.

Diverging attributes have a midpoint (or zero value), where values greater than the midpoint are positive, and values less than the midpoint are negative. Examples include temperature and altitude.

Cyclic attributes repeat over a set period. Examples include time of day, months of the year, seasons, and compass directions.

→ Sequential



→ Diverging



→ Cyclic



Ordered attributes are divided into two types: ordinal and quantitative.

ORDINAL ATTRIBUTES

Ordinal attributes have an implied order, but numerical calculations, such as subtracting or computing a mean, are not possible.

Examples of ordinal attributes include:

- Clothing sizes (S, M, L, XL)
- Rankings in a competition (1st, 2nd, 3rd)
- Grades (HD, D, C, P, N)
- Days of the week (Mon, Tue, etc.)
- Highest education (high school, undergraduate, graduate, PhD)
- Customer satisfaction levels (dissatisfied, neutral, satisfied)

QUANTITATIVE ATTRIBUTES

Quantitative attributes are real numbers and integers, allowing arithmetic operations, such as calculating the mean value.

Examples of quantitative attributes include:

- House prices in AUD
- Temperature in degrees Celsius
- Geographic positions in latitude and longitude

Quantitative attributes can be converted into ordinal attributes. For example, quantitative marks over 80% can be mapped to the HD ordinal attribute, marks between 70% and 80% to D, and so on. However, the inverse conversion from ordinal attributes to quantitative attributes is not possible.